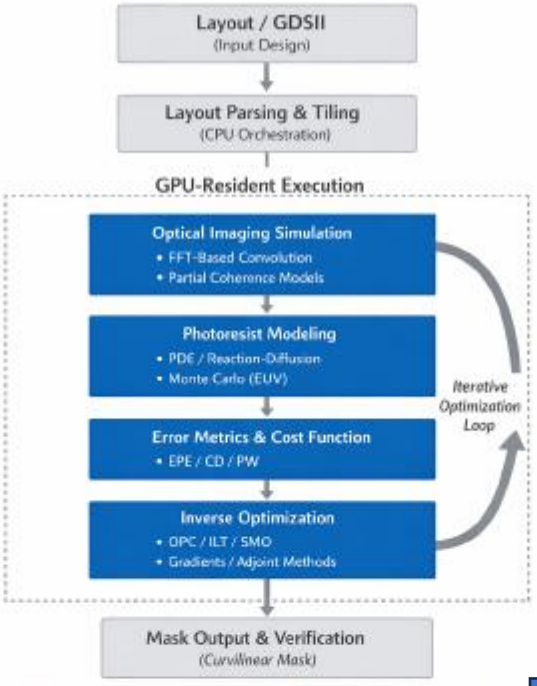


Fig. 1 Conceptual GPU-centric computational lithography pipeline. Optics modeling, error evaluation, and inverse optimization are executed as a fully workflow, minimizing data movement and enabling efficient single-GPU, m scale execution.

You may elaborate more for clarity.

Full pipeline GPU residency still involves CPU-GPU data transfer but it hides the latency by separating compute and transfer stages and pipelining them.

Equally important is how GPU acceleration is integrated int raphy workflow. Early approaches often offloaded isolated while retaining control logic and intermediate data on suffer from significant overhead due to repeated h vice da recent frameworks emphasize full-pipeline GPU idency, where optical simulation, resist modeling, and inverse optimization remain device-resident across iterations. This design strategy reduces communication overhead, improves energy efficiency, and enables more effective multi-GPU scaling on cluster architectures [12, 16].

Motivated by these developments, this paper presents a comprehensive survey of GPU-accelerated computational lithography. We provide a taxonomy spanning application domains, numerical methods, and system architectures; synthesize trends across academic research and industrial deployment; and analyze key challenges related to scalability, memory behavior, numerical precision, and reproducibility. Finally, we outline a roadmap for future research at the intersection of GPU-accelerated clusters, high-performance computing, and next-generation semiconductor manufacturing. Unlike prior surveys that focus primarily on lithography physics or OPC algorithms,

22 nm nodes. As OPC derived tools enter the stage
standpoint, their use model is being greatly broadened
out flow, upstream, directly integrated into physical de
awareness into the physical design environment, media
tools and flows, is driving various forms of manufactura
tation: from Restricted Design Rules and Flexible Des
Fabrics. As new lithography solutions, such as immersi
have to be deployed within a complex technology framework, t
also examines
the trend towards "layout design regularization" and its im
ions for patterning
and next generation lithographies.

This paragraph seems to be written from the perspective of author of [26] Pang et.al. Change the wording to describe your take on author's findings

Inverse Lithography Technology (ILT) is a rigorous approach to determine the mask shapes that produce the desired on wafer results [26]. In this paper, we briefly describe an image (or pixel)) based implementation of ILT in comparison to OPC technologies, which are usually edge based. Such implementation is more computationally scalable and avoids laborious segmentation script writing, which becomes more complex for newer generations because of complicated proximity effects. In this paper, we will give an overview of ILT, present some simulation and wafer examples to demonstrate the benefit of ILT, clarify common myths about ILT, discuss and show examples to illustrate the impact in every step of the mask making process. Specifically, studies done with several leading mask shops around the world on mask manufacturability (including data fracturing, writing strategy and writing time, mask inspection), will be shown.

As Moore's law has marched forward, progressively shrinking chip designs at a consistent pace, the manufacturing of those chips has raced to keep up. Through advances in lithography hardware and software, we now are in the realm of the single digit nanometer design nodes at leading chip foundries. Cecil et al.[2] reviews the method of ILT, which is the preferred computational method for photolithography. Indeed photolithographic masks were the first metasurfaces, and still the most important metasurface, economically, used in memory chips, storage, and microprocessors.
Moreover, photolithographic mask designers were t
matical optimization in optics, creating ILT, specif
metrics, and using the mathematical machinery of
and adjoints to drive the mask design process.

Here the start of paragraph was good but then the perspective changed from authors to Chen et. al. Please write you take on the paper by Chen et.al.

Chen et al. [3] talk about how Neural network bas
(NNILT) has been used to improve the computatio
optimization for advanced photolithography. NNIL
and its performance is affected by the quality of labels.
ult for NNILT to
achieve high performance and extrapolation ability fo
optimization without
using labels. Here, we propose a label free NNILT (LF NNILT), which is implemented completely without labels and greatly improves the printability of the target layouts and the manufacturability of the synthesized masks compared to the traditional ILT. More importantly, the optimization speed of LF NNILT is two orders of magnitude faster than the traditional ILT. Furthermore, LF NNILT is simpler to implement and can achieve better solvers to support the development of advanced lithography.

GPUs in a cluster. Conversely, divergence-heavy kernels limit achievable speedup and reduce performance per watt, undermining the benefits of GPU acceleration. As a result, modern GPU-accelerated lithography frameworks increasingly favor pixel-based and grid-based formulations over control-intensive approaches, structuring entire optimization pipelines to remain device-resident and SIMT-friendly [4, 29].

In summary, SIMT execution aligns well with lithography workloads due to their inherent pixel-level parallelism and grid-based numerical structure. Achieving high performance requires careful kernel design that maps FFT grids and resist stencils cleanly onto warps while minimizing divergence introduced by conditional logic. These considerations are fundamental to realizing scalable, energy-efficient GPU acceleration in computational lithography.

4.2 Memory Hierarchy and Data Movement

Memory behavior is a dominant performance and energy-efficiency factor in GPU-accelerated computational lithography. Unlike compute-bound kernels with minimal data reuse, lithography workloads repeatedly a
sets—including mask grids, pupil functions, optical trans
variables—across thousands of iterations. As a result, ef
memory hierarchy is essential for sustaining high throug
bottlenecks [9, 10].

FFT-based optical simulation is particularly sensitiv
Large two-dimensional FFTs require multiple passes o
computation with global memory reads, writes, and data req ges. When
data are laid out contiguously and accessed in a coalesced ma PUs can exploit
their high global memory bandwidth to achieve near-peak ughput. However, sub-
optimal layouts or frequent data transposition
by increasing memory traffic and cache misse [9, 10]. Document A highlights that
full-chip lithography simulations often become
compute-bound, especially when operating on large frequency-domain grids.

Please clarify what is "Document A"
Is it [9]? Maybe something like following:
Owens et. al. [9] highlight that…

Resist modeling introduces additional memory pressure. Reaction–diffusion solvers maintain multiple state variables per grid point and require repeated stencil-based updates that read from neighboring locations. While these access patterns exhibit spatial locality, their performance depends strongly on how intermediate fields are staged within the memory hierarchy. GPU-accelerated lithography studies show that retaining frequently accessed resist fields in on-device memory across iterations substantially reduces global memory traffic and improves overall efficiency [4].

Data movement between the host and device represents one of the most significant sources of overhead in early GPU-based lithography implementations. Initial approaches often transferred intermediate results—such as aerial images or resist profiles—back to the CPU between simulation stages or optimization iterations. These transfers incur high latency and energy cost, quickly offsetting computational gains from GPU acceleration [9]. More recent frameworks emphasize minimizing host–device communication by keeping the entire lithography pipeline device-resident, including optical simulation, resist modeling, and inverse optimization loops [4, 29].

increasingly limit yield at advanced technology node[illegible]. As discussed in Document A, resist modeling is computationally demanding due to the use of stencil-based reaction–diffusion solvers and stochastic Monte Carlo simulation[illegible]icularly in extreme ultraviolet (EUV) lithography where low photon cou[illegible] effects [6, 18].

Again, please clarify what is "Document A"

From a computational perspective, resist simulat[illegible] metic intensity and substantial memory pressure. R[illegible] repeated updates over dense spatial grids, while Monte Carlo approaches demand large numbers of statistically independent samples to obtain stable variability estimates. GPU acceleration in this category exploits strong spatial locality and massive parallelism across grid points or stochastic samples, enabling efficient evaluation of variability metrics that would otherwise be prohibitive on CPU-based systems. These characteristics make resist simulation particularly well suited to SIMT execution and multi-GPU scaling.

Resist modeling is often invoked not only as a standalone analysis tool but also as a tightly coupled component within OPC and ILT workflows, where it must be evaluated repeatedly across optimization iterations and process conditions. In such settings, retaining resist state variables and intermediate fields in device memory is critical for minimizing data movement overhead and sustaining throughput. GPU-resident execution has therefore become a key design principle for scalable resist simulation, especially when integrated into inverse lithography pipelines that require thousands of resist evaluations [4].

5.1.5 Electromagnetic (EM) and Extreme UltraViolet (EUV) Modeling

Rigorous electromagnetic simulation becomes necessary when scalar diffraction models are no longer sufficient, such as in high–numerical-aperture (high-NA) and extreme ultraviolet (EUV) lithography, where vector effects, polarization, shadowing, and three-dimensional mask topography significantly influence image formation and pattern fidelity [7, 19]. In these regimes, EM solvers resolve Maxwell's equations to accurately capture near-field interactions and mask-induced effects that cannot be represented by FFT-based imaging models.

Despite their accuracy, EM-based lithography models are substantially more computationally expensive than scalar optical simulations. They involve large-scale field updates, dense linear algebra, and fine spatial discretization, leading to high memory footprints and long runtimes. Consequently, EM solvers are typically applied selectively in practice—for example, to analyze critical features, validate simplified optical models, or calibrate reduced-order representations used in full-chip simulation flows [36]. This selective use reflects a pragmatic balance between physical fidelity and computational feasibility.

GPU acceleration in EM and EUV modeling primarily targets dense numerical kernels such as field propagation, matrix operations, and domain-decomposed updates. While GPUs can provide significant speedups for these kernels, scalability is often constrained by memory capacity, irregular data access patterns, and inter-device communication overhead, particularly in three-dimensional simulations. As a result, EM

that shaped the transition from early academic exploration to large-scale industrial adoption.

- **2005 - 2015 Early Academic Exploration of GPU-Accelerated FFTs**: The initial phase of GPU adoption in computational lithography focused on accelerating FFT-based optical simulations. Academic studies during this period demonstrated that GPUs could significantly outperform CPUs for large-scale Fourier-domain imaging, establishing the feasibility of GPU acceleration for core lithography kernels. These early efforts primarily targeted standalone optical simulations and proof-of-concept acceleration rather than full lithography workflows.
- **2015 - 2020 Emergence of OpenILT and Experimental GPU Acceleration**: The introduction of research platforms such as OpenILT marked a shift toward system-level experimentation with GPU-accelerated inverse lithography. During this period, GPUs were increasingly used to accelerate not only optical imaging but also optimization loops and gradient computation. Although these efforts demonstrated strong scalability potential, GPU acceleration remained largely confined to academic and experimental settings, with limited penetration into production EDA tools.
- **2020 - 2023 Initial GPU Integration in Commercial Tools**: Between 2020 and 2023, major commercial lithography tools—including Synopsys S-Litho—began integrating GPU support for selected computational modules. This phase represented a transitional period in which GPU acceleration moved from research prototypes into production environments. However, most deployments adopted hybrid CPU–GPU architectures, accelerating isolated kernels while retaining CPU-centric workflow control.
- **2023 - 2025 GPU-First Platforms and Curvilinear ILT**: The launch of NVIDIA cuLitho marked a significant inflection point by demonstrating a fully GPU-resident lithography pipeline capable of delivering up to 40× speedups over traditional CPU-based flows. In parallel, the introduction of curvilinear ILT tools enabled efficient synthesis of advanced mask geometries required at state-of-the-art nodes. Together, these developments signaled a shift toward GPU-first architectures designed explicitly for large-scale inverse optimization and cluster execution, rather than incremental acceleration of legacy flows.

D2S Inc. introduced TrueMask ILT in Sep, 2019 which is also a GPU-First Curvilinear ILT. So you need to update the timeline here

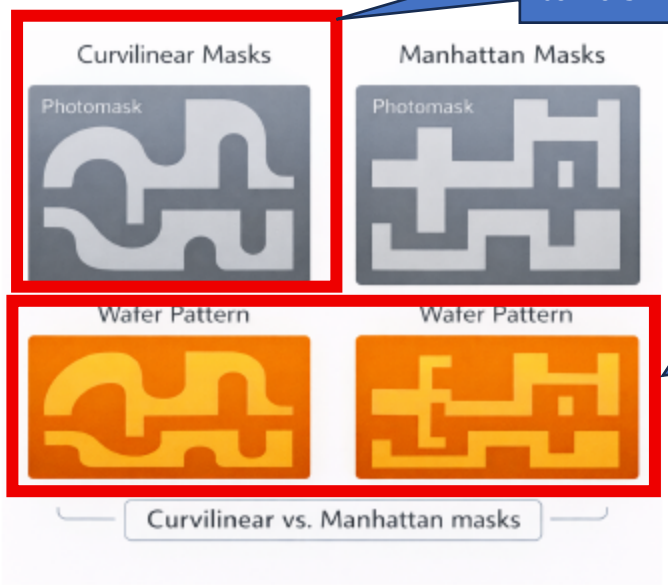
6.2 Indu[illegible]tforms

I[illegible]ctability, and tight integ[illegible]ows. GPU acceleration in p[illegible]ully engineered pipelines, optimized numerical libraries, and an emphasis on sustained throughput rather than peak kernel performance.

6.2.1 NVIDIA cuLitho

NVIDIA's cuLitho platform represents a GPU-first redesign of computational lithography for production-scale deployment [14]. Rather than accelerating isolated kernels, cuLitho integrates optical simulation, resist modeling, and inverse optimization into a unified, GPU-resident pipeline. By leveraging optimized FFT libraries, custom CUDA

Curvilinear photomasks should not have 90 degree corners



Images seem incorrect if they represent patterns achieved on wafer. Does this represent wafer target patterns? Target patterns may have 90 degree corners. Please clarify what the intent is.

Fig. 3 Comparison of traditional Manhattan mask geometries and curvilinear mask representations, along with their corresponding wafer patterns. Curvilinear masks enable improved pattern fidelity and process window by better compensating for diffraction and process effects at advanced technology nodes.

More recent advances integrate both physics-based optimization and data-driven techniques. Cao et al. explored the application of generative AI in conjunction with ILT to enable efficient synthesis of curvilinear mask patterns, illustrating how modern GPU platforms can support hybrid optimization pipelines [29]. Yang et al. proposed a GPU-accelerated ILT algorithm that improves contour quality, process window robustness, and curvilinear mask precision, reporting speedups of up to 40× compared to CPU-based ILT methods [21]. In addition, Yang et al. discussed the fundamental limitations of conventional OPC and emphasized how ILT and curvilinear masks address critical fidelity challenges in advanced wafer fabrication [21]. Beyond ILT, GPU acceleration has also been applied to related patterning problems such as multiple-patterning layout decomposition, further underscoring the importance of GPU-based optimization for advanced mask synthesis [48].

Together, these innovations demonstrate that curvilinear ILT is both a major driver and a beneficiary of GPU acceleration. The combination of FFT-heavy imaging, gradient-based optimization, and dense mask representations makes GPU-dominant and multi-GPU architectures a necessity rather than an optimization choice.

6.5 Lithography Toolchain Integration

While GPU acceleration has transformed lithography simulation and optimization, its full impact is realized only when integrated across the broader lithography toolchain. This includes compatibility with different exposure technologies and downstream

mask writing systems that impose additional performance and data management requirements.

6.5.1 Electron Beam and Deep Ultraviolet Lithography Context

Electron beam (e-beam) lithography and deep ultraviolet (DUV) lithography continue to play important roles in both mask fabr
Lin et al. described use cases for e-beam and I
manufacturing settings, highlighting their relevar
prototyping [49]. Tseng et al. reviewed the evolut
an emphasis on nanometer-scale device fabrica
resist processes, and pattern control that enable

Despite their high resolution, traditional e-b
itations that restrict scalability for volume ma
multiple electron beam techniques aimed at im
lithography, noting their potential impact on dire
[51]. These studies provide important context f
lithography, coupled with GPU acceleration, is e
and data volume associated with modern patter

6.5.2 Integration with Multi-Beam M

Multi-beam mask writers (MBMWs) represen
translating curvilinear ILT designs into manufa
thousands of parallel electron beams, MBMW
time while maintaining the precision required
EUV lithography. However, the adoption of cur
challenges related to data volume, throughput, a

Shamoun et al. discussed the evolving requi
writers by EUV lithography, emphasizing the
improved critical dimension uniformity, reduced
induced defects [36]. They also highlighted the gr
data volumes associated with curvilinear ILT and the im
between mask writers and computational lithograp

Yasuda et al. reviewed the development of NuFlare Technology's multi-beam mask writing systems, describing architectural features such as high beam current density, pixel-level dose correction, and charging-effect mitigation designed for high-volume manufacturing of leading-edge masks [52]. These advances underscore the tight coupling between GPU-accelerated ILT, curvilinear data representations, and downstream mask writing infrastructure.

From a system perspective, effective integration between GPU-based lithography computation and multi-beam mask writers is essential for realizing the benefits of curvilinear ILT in production. Managing data growth, preserving pattern fidelity, and sustaining throughput across the toolchain require coordinated advances in algorithms, system architecture, and hardware. As such, MBMW integration represents a key convergence point where GPU-accelerated computation directly influences manufacturing feasibility.

Since you are talking about MBMWs and GPU usage you may consider mentioning the following:

NuFlare multibeam mask writers also use GPU accelerated Mask Process Correction (MPC) software called PLDC. Here GPU acceleration enables inline MPC for photomasks that adds zero turnaround time for MPC.

References:

Abhishek Shendre, Nagesh Shirali, Aki Fujimura, "Making curves manufacturable with O(p) computing: a sampling theory approach," Proc. SPIE 13687, Photomask Technology 2025, 136870H (6 November 2025); https://doi.org/10.1117/12.3072119

Noriaki Nakayamada, Haruyuki Nomura, Yasuo Kato, Kenichi Yasui, Abhishek Shendre, Nagesh Shirali, Yukihiro Masuda, Aki Fujimura, "Curvilinear mask process correction embedded on multi-beam mask writer," J. Micro/Nanopattern. Mats. Metro. 23(1) 011206 (14 February 2024) https://doi.org/10.1117/1.JMM.23.1.011206